

Supplementary Material for the paper “SimFusion: Multi-Fidelity Simulation-based Testing of Automated Driving Systems”

This supplementary material provides visualizations and extended results for the SimFusion framework described in the main paper. The experiments have been conducted on a 16 Core CPU PC with NVIDIA GeForce RTX 4090 and 32 GB RAM.

A. Case Studies

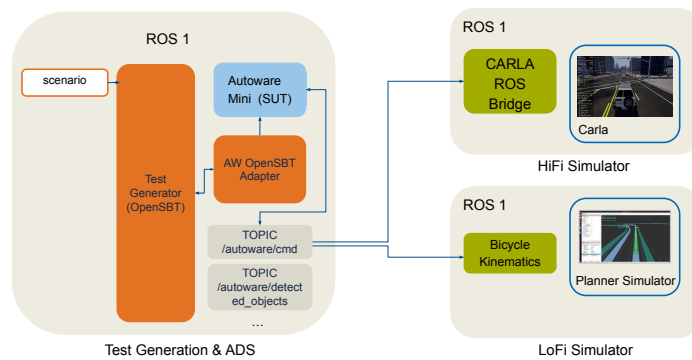


Fig. 1: Autoware case study deployment setup. All systems, i.e. SimFusion Autoware Mini with Carla, and Autoware Mini with LoFi Simulator are deployed in designated docker containers. Test execution is enabled by sending command via ROS topics to the SUT and consuming telemetry data available in the ROS network.

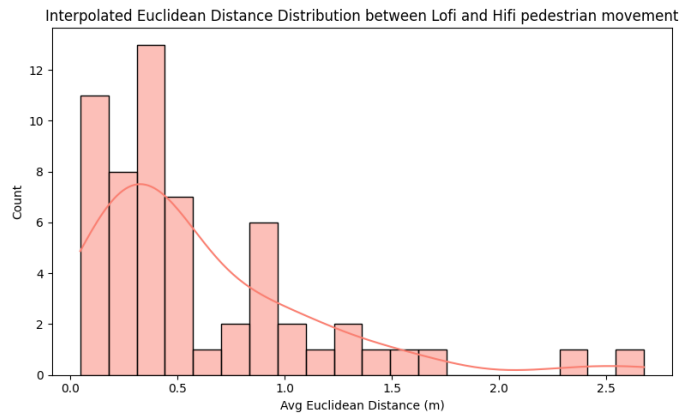


Fig. 2: (Autoware) Scenario alignment validation by measure the average Euclidean distance between a pedestrian simulated in LoFi and HiFi for one and the same scenario sampled over 100 scenarios.

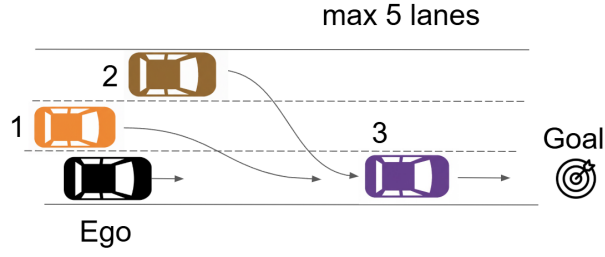


Fig. 3: Overview of scenario modelled in the FrenetiX case study. The scenario is defined by 11 parameters, among those the number of lanes and the position of surrounding vehicles and the goal is controlled.

B. Flakiness Analysis (Autoware)

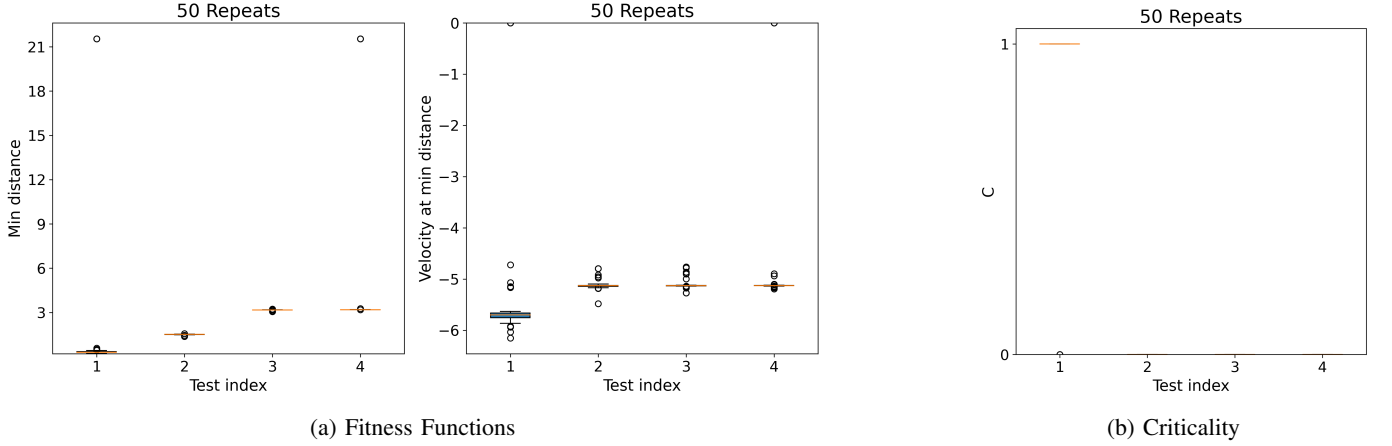


Fig. 4: (Autoware) Fitness variation by reexecuting randomly generated scenarios 50 times in the Carla-based HiFi simulator. Both distance fitness and velocity fitness variation are low and no variation of the tests outcome (Figure b)) can be observed indicating low simulation flakiness.

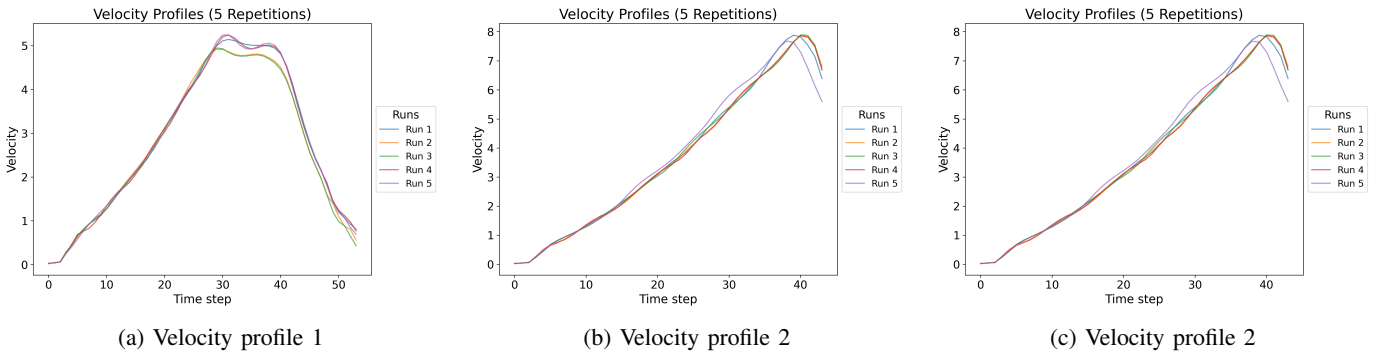


Fig. 5: (Autoware) Overview of velocity profiles from reexecution of random scenario 5 times. The profiles reveal no significant deviation as indicated by the low fitness score variation in Figure 4.

C. Effectiveness Test Results

TABLE I: (RQ2 Effectiveness) Statistical comparison of SimFusion against baselines using failure ratio. Statistical significance is assessed using Wilcoxon rank-sum test ($\alpha = 0.05$), and effect size is reported using the Vargha–Delaney statistic $\hat{A}_{12}$.

Case Study	Comparison	p-value	$\hat{A}_{12}$
Autoware AEB	SimFusion vs LoFi	0.00250	1.000 (Large)
	SimFusion vs HiFi	0.00320	0.986 (Large)
	SimFusion vs Surrogate	0.29437	0.611 (Small)
Autoware CUT-IN	SimFusion vs LoFi	0.00139	1.000 (Large)
	SimFusion vs HiFi	0.01299	0.889 (Large)
	SimFusion vs Surrogate	0.00108	1.000 (Large)
Frenetix	SimFusion vs LoFi	0.00246	1.000 (Large)
	SimFusion vs HiFi	0.00250	1.000 (Large)
	SimFusion vs Surrogate	0.00108	1.000 (Large)

D. Efficiency Test Results

TABLE II: (RQ3 Efficiency) Statistical comparison of SimFusion against baselines using time per failure. Statistical significance is assessed using Wilcoxon rank-sum test ($\alpha = 0.05$), and effect size is reported using the Vargha–Delaney statistic $\hat{A}_{12}$.

Case Study	Comparison	p-value	$\hat{A}_{12}$
Autoware AEB	SimFusion vs LoFi	0.00108	1.000 (Large)
	SimFusion vs HiFi	0.00108	1.000 (Large)
	SimFusion vs Surrogate	0.00758	0.917 (Large)
Autoware CUT-IN	SimFusion vs LoFi	0.00136	1.000 (Large)
	SimFusion vs HiFi	0.03254	0.833 (Large)
	SimFusion vs Surrogate	0.00250	1.000 (Large)
Frenetix	SimFusion vs LoFi	0.00246	1.000 (Large)
	SimFusion vs HiFi	0.00250	1.000 (Large)
	SimFusion vs Surrogate	0.40443	0.556 (Negligible)

E. Implementation Details

We implemented SimFusion in Python on top of OpenSBT [1], a modular framework for search-based testing of autonomous driving systems used in similar research, which is publicly available [2]. For Autoware, CARLA, the ROS-based simulator, and Autoware Mini are deployed using Docker containers. Scenarios in CARLA are executed via CarlaRunner [1], while LoFi simulations are triggered through ROS message exchange [3]. For the Frenetix case study, BeamNG runs on Windows, while CommonRoad and Frenetix are deployed in WSL due to dependency constraints [4]. To ensure robustness of long-running experiments, we implement a simulator restart mechanism to handle occasional execution interruptions.

REFERENCES

- [1] L. Sorokin, T. Munaro, D. Safin, B. H.-C. Liao, and A. Molin, “OpenSBT: A modular framework for search-based testing of automated driving systems,” New York, NY, USA, p. 94–98, 2023. [Online]. Available: <https://github.com/opensbt/opensbt-core>
- [2] “Replication package.” 2026. [Online]. Available: <https://anonymous.4open.science/r/simfusion-C957/README.md>
- [3] UT-ADL, “Autoware Mini: a minimalistic python-based autonomy software,” https://github.com/UT-ADL/autoware_mini, UT Autonomous Driving Lab, 2023, accessed: 2026-02-24.
- [4] M. Kaufeld, K. Moller, A. Gambi, P. Arcaini, and J. Betz, “Multidrive: A co-simulation framework bridging 2d and 3d driving simulation for av software validation,” in *IEEE 28th International Conference on Intelligent Transportation Systems (ITSC 2025)*, Nov. 2025.